BITS SCHOOL OF MANAGEMENT
Term 6, Block 33, 2025 - 26
Course Outline

Course Title: Fintech

Faculty: Prof Pulak Ghosh

Dates: Feb 2nd to Feb 15th, 2026

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Office Hours: By prior appointment only

Course Description:

Digital innovations have reshaped the finance industry, leading to the emergence of Fintech. Fintech has changed how financial information is disseminated, processed and analyzed. Individual investors and financial institutions who are able to leverage these new technologies are having a leading edge. In this course we will explore the economic forces reshaping financial services and some key implications for building competitive advantage and designing sound regulations. We will also learn broad introduction to Financial Technology, influence of emerging technology in Consumer financial decision. Students will learn the technology behind payments, loans, trading, and asset management as well as applications of machine learning, blockchain technology for finance and transactions. Using both case studies and analytical templates, students will get an understanding of the challenges and opportunities in this space. Innovation in Fintech combined with novel business models, advanced tech, data and networks leading companies in the financial services industry. How FinTech's and Banks are reshaping the financial services Ecosystem will help you learn and adapt, providing actionable, strategic insights for decision making in the fast changing financial services sectors.

Learning Objectives / Outcomes

The course is designed with the following specific objectives and learning outcomes:

a. Understand the impact of disruptive tech in financial services
b. Learn how start up, fintech and banks responding to innovation and disruption in the space
c. Understand the effect of ChatGPT on financial services
d. Understand blockchain and cryptocurrency
e. Explore strategies for managing innovation and how to embrace it

Course Materials: Will be shared on LMS

Assessment: Maximum Marks 100

It will be a quantitative grading with three components:

a) Class participation 20%,
b) Mid-term 40%; and
c) Group project 40%

Groups of 5-6 students. Students can pick their own groups.

Mid-term Exam: scheduled for 7th February

Project: A small group of 5-6 students will be created. They have to come up with a new/fresh start-up idea on fintech and defend their idea on 4 parameters (a) how it will disrupt the market (b) how and when they think it will make money (c) what is the USP of the idea (d) how would they pitch it to VC for funding.

POLICY:

Class Policies

We will start classes on time.

If you are late, the instructor has the choice of not allowing you to attend or creating creative punishments for tardiness.

Attendance Policy

The attendance policy as laid out by the Programme Office applies. Please refer to your student handbook for details.

Honour Code

BITSoM takes the honour code seriously, and so should you. The student handbook contains details on the honour code policy at BITSoM. You are expected to abide by an honour code and follow a culture of honesty. Please ensure that you read the relevant section(s) in the handbook. Feel free to ask us in case you have any questions.

Coding scheme for all course work:

Code	Nature of Course work Discussion		Nature of Reference Material	
	General Discussions	Specific Discussions	External Material	Case/Problem Solutions
AC-I	Not Allowed	Not Allowed	Not Allowed	Not Allowed
AC-IIa	Allowed	Not Allowed	Not Allowed	Not Allowed
AC-IIb	Not Allowed	Not Allowed	Allowed	Not Allowed
AC-IIIa	Allowed	Allowed	Not Allowed	Not Allowed
AC-IIIb	Allowed	Not Allowed	Allowed	Not Allowed
AC-IIIc	Not Allowed	Not Allowed	Allowed	Allowed
AC-IV	Allowed	Allowed	Allowed	Not Allowed
AC-V	Allowed	Allowed	Allowed	Allowed

Component	Honor Code
Class participation	AC- V
Mid-term	AC-I
Group project	AC-IIb

Session Schedule	
Module & Sessions	**Topic**
Session 1	**Topic:** Overview of Fintech: Digital Transformation We discuss the broad spectrum Fintech, applications, debate and changes in financial service industry. Will discuss broad examples of success and failures of various applications. **Readings:** • The Economist, To Do With the Price of Fish (http://www.economist.com/node/9149142) • Philippon, Thomas. 2018. The Fintech Opportunity. NBER Working Paper. http://pages.stern.nyu.edu/~tphilipp/papers/FinTech.pdf • Wharton, Why Fintech is Disrupting Traditional Banking (2019) • To Fintech and Beyond (Review of Financial Studies) • Value of FinTech (https://assets.kpmg/content/dam/kpmg/uk/pdf/2017/10/value-of-fintech.pdf) • The Evolution of Fintech - The New York Times (https://www.nytimes.com/2016/04/07/business/dealbook/the-evolution-of-fintech.html)
Session 2	**Topic:** Enablers of Digital Economy (Tech part of Fintech) We will cover the following topics: Big data, value of data, ML, AI, chatbots, ChatGPT, Generative AI, LLM in finance, agentic AI **Readings:** • Dhar V., When to Trust Robots with Decisions and When Not To, Harvard Business Review, May 2016. • Christensen, Clayton M., Raynor, Michael E. and McDonald, Rory. (2015). What is Disruptive Innovation? Harvard Business Review.

	• State of AI report -2022 (stateof.ai by Nathan Benaich and Ian Hogarth • https://bfsi.economictimes.indiatimes.com/news/financial-services/how-chatgpt-the-new-ai-wonder-may-transform-bfsi/97083280 • ONDC strategy paper: https://www.medianama.com/wp-content/uploads/2022/03/ONDCStrategyPaper.pdf • JPMorgan: Big Data and AI Strategies: Machine Learning and Alternative Data Approaches to Investing
Session 3-5	**Topic:** Fintech applications as disruptions in business We will cover the following topics: • Fintech for lending: [Consumer finance, B2C, MSME, BNPL] • P2P lending, BNPL • Robo-Advisor • Competition between bank and fintech **Readings:** • Christopher Nicholls, Open Banking and the Rise of Fintech: Innovative Finance and Functional Regulation (2019). • The State of Digital Lending - American Bankers Association https://www.aba.com/Products/Endorsed/Documents/ABADigitalLending-Report.pdf • Data and welfare in credit Markets (https://www.nber.org/papers/w30235)
Session 6-7 **Session 8**	**Payment systems and open banking** We will explore various payment systems and pros and cons of those Ref: https://www.nber.org/papers/w33259 https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5147405 **Topic:** Introduction to Blockchain, Crypto and CBDC We will cover the following topics: • Blockchain uses cases and digital assets • CBDC **Readings:** • Halaburda, Haeringer, Gans and Gandal (2020), "Microeconomics of cryptocurrencies", NBER working paper

	• Orcutt (2017), "Wait, Bitcoin Just Did What?," MIT Technology Review • Hamacher (2019), Why Bosch is jumping on the Ethereum blockchain • Howell, Niesser and Yermack (2019), "Initial Coin Offerings: Financing Growth with Cryptocurrency Token Sales," NYU working paper • Dale (2019), "Decentralized Airbnb Starts Charging Fees as ICO Model Falters," Coindesk • Ertz and Boily (2018), "The Rise of the Digital Economy: Thoughts of Blockchain Technology and Cryptocurrencies for the Collaborative Economy," University of Quebec working paper • Leiponen (2020), "Platform ecosystem in the dApp economy," Cornell University working paper
Session 9	**Topic:** Big Issues in Fintech- Privacy, Discrimination and Bias We will cover the following topics: • Data Privacy • Discrimination by AI • Algorithmic Aversion • ML bias **Readings:** • Big data, financial services and privacy. The Economist, February 9, 2017 • Robinhood concerns: http://fortune.com/2018/12/16/robinhood-checking-savingsregulatory-concerns/ • Nik Milanovic, Now more than ever we need fintechs to lead on consumer transparency (2020). • Johannes Ehrentraud & Denise Garcia, Managing the winds of change: policy responses to fintech (2020). • Robert Barlett, Adair Morse, Richard Stanton & Nancy Wallace, Consumer-Lending Discrimination in the Fintech Era (2019). • Jim Saksa, Rise of fintech weakens law to prevent lending discrimination (2019).
Session 10	Project Presentation